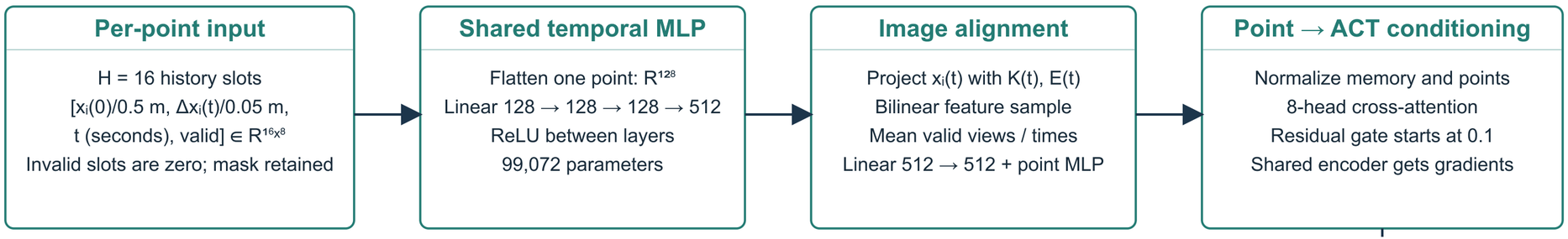
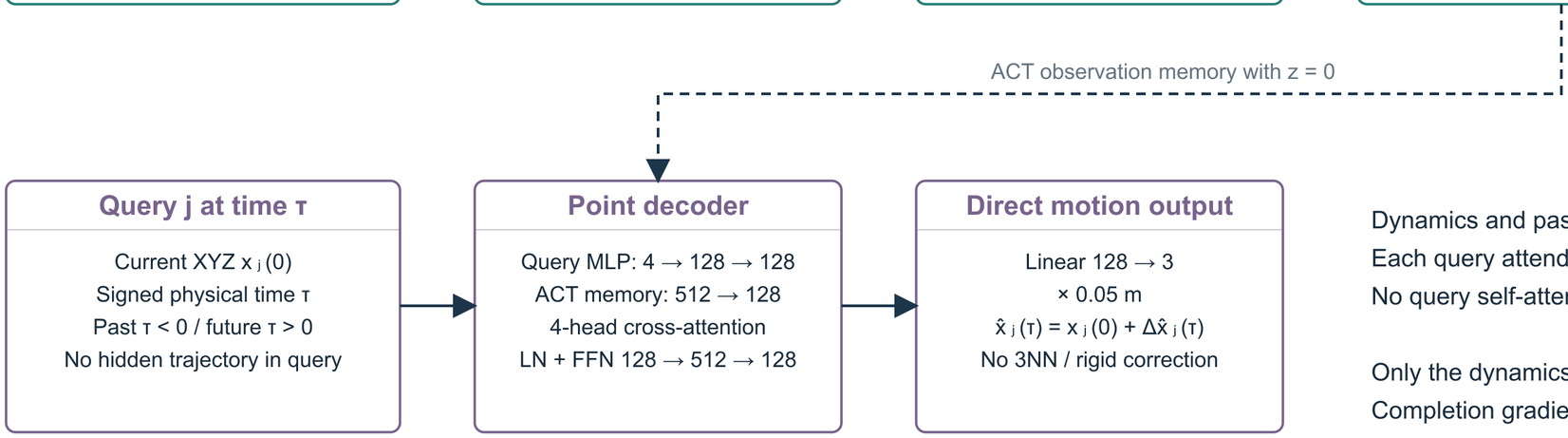
Inside the point branch

Causal, identity-preserving inputs; direct displacement prediction; no nearest-neighbor interpolation prior



K points have no learned ID or order embedding. Changing point order only changes token order.



Dynamics and past completion use separate heads.
Each query attends to the same observed memory.
No query self-attention: query chunks preserve outputs.

Only the dynamics head also reads committed actions.
Completion gradient is detached or explicitly norm-capped.

Controlled comparison: stationary tracks vs observed tracks

Both branches use the same current anchors, timestamps, validity, visual backbone and parameter budget. Stationary XYZ is applied before image gathering.